Fast Maximum Likelihood Estimation for 1D Gaussian Splits

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***Abstract*—Gaussian Mixture Models (GMMs) are widely used to approximate complex probability density functions. While the EM algorithm is the standard method to estimate GMM parameters, determining the number of mixture components remains a challenging model selection problem. Incremental approaches address this issue by gradually evolving a set of components, where a key step is the split operation: choosing how to divide a Gaussian component into two. However, the split problem is ill-posed, since it admits infinitely many solutions, and the choice of split strategy can critically affect the convergence and accuracy of subsequent EM iterations. In this work, we focus specifically on the split function rather than on a complete estimation algorithm. We propose the 1-dimensional Gaussian Pair Mixture (GPM1D) algorithm, a non-iterative maximum-likelihood method to split a Gaussian component. The idea is to project the component onto a one-dimensional subspace aligned with the intended split direction, and then efficiently fit a two-component Gaussian mixture in this space. The method is deterministic, parallelizable, and computes the global optimum (up to discretization) in fixed time, enabling a controllable trade-off between accuracy and runtime. We present the method and compare it experimentally against EM-based splits across different data sizes. Because our contribution is limited to the split step, GPM1D can be directly integrated into various incremental or greedy GMM estimation frameworks.**

***Index Terms*—Gaussian Mixture Models, Greedy Methods, Ill-posed Problem. Splitting components, Unsupervised learning, Unsupervised clustering**

I. Introduction

Unsupervised clustering classifies data into groups based on redundancies in the sample, without external supervision. A common way to model such data is by fitting a Gaussian Mixture Model (GMM), which provides a compact probabilistic representation of stochastic phenomena [10]. Since many real-world variables can be approximated by Gaussian distributions, clustering under the GMM assumption reduces to estimating the number of mixture components and their parameters.

The Expectation–Maximization (EM) algorithm is the standard approach for parameter estimation in mixture models [3], [8], [9], [11]. Although EM is attractive for its generality, it has some well-known drawbacks: (i) it requires the number of components to be fixed a priori, (ii) it is sensitive to initialization, and (iii) it converges only to local optima [10], [21]. Determining the number of mixture components is particularly challenging [4], and several approaches attempt to address it by jointly searching over both model order and parameters.

Incremental methods are a widely adopted class of solutions. They start with a coarse mixture model and progressively refine it by splitting existing components into two, followed by EM re-estimation [3], [21], [10], [7]. The process stops when a model selection criterion such as AIC [13], BIC [14], MDL [12], or MML [20] no longer justifies further refinements.

A crucial step in these approaches is the split operation. Splitting a Gaussian component into two new ones is inherently ill-posed, since infinitely many solutions exist [23]. Common strategies rely on heuristics, such as random perturbations of parameters [16], higher-order moments [18], [5], moment matching with prior assumptions [22], or axis-aligned splits [1]. However, because EM is initialization-sensitive, suboptimal splits often lead to slower convergence and poorer local optima.

In this work, we focus exclusively on the split function rather than on a complete estimation procedure. Our goal is to design a split rule that places the two new components as close as possible to the optimal solution, thereby improving convergence speed and accuracy in subsequent EM steps. To this end, we first project the data onto a one-dimensional subspace aligned with the split direction, which reduces the computational complexity [17]. Then, a global maximum-likelihood optimization fits two Gaussian components in this subspace, providing a principled and deterministic split.

In summary, while much prior work has focused on full GMM estimation procedures, our contribution targets the specific but critical subproblem of splitting a Gaussian component. We introduce the 1-dimensional Gaussian Pair Mixture (GPM1D) method, a deterministic and non-iterative split function based on maximum likelihood in a projected subspace. The remainder of the paper is organized as follows. Section II describes the related work. Section III describes the proposed method in detail. Section IV presents experimental results comparing GPM1D with EM-based splits. Section V discusses implications and possible integrations into incremental GMM frameworks. Finally, Section VI concludes the paper and outlines directions for future work.

II. Related Work

In this section we review some past work on splitting based approaches for GMM optimization. In 1999, Vlassis and Likas [19] proposed an algorithm that selects the mixture component with fourth order moment (*kurtosis*) more distinct than that of a regular Gaussian, and splits it into two new ones with identical weights and variance identical to the original component. Then, the EM algorithm is run on the new mixture until convergence. In 2007, Constantinopoulos and Likas [1] splits are optimized using a local Variational Bayes Approach (VBA). This local VBA optimizes the parameters of the new components by maximizing the marginal likelihood of the data with a particular choice of priors on the parameters of the components. After convergence, the outcome of the local VBA may be one of three: (i) the two components converge to non-singular configurations of the covariance matrix and a better fit to the data (by construction of the VBA method), (ii) one of the components converges to a singular configuration, or (iii) both components converge to singular configurations. Cases (ii) and (iii) are degenerate and are considered failures of the split. In 2011, Dasgupta [2] has demonstrated the advantages of using low-dimension projections in Gaussian Mixture Estimation using a variant of the EM algorithm that starts on a low-dimension subspace of the original data. In 2024, Greggio and Bernardino [6] developed a solution that projects the candidate component to split into the eigenvector direction corresponding to the maximum eigenvalue of the covariance matrix of the candidate component, and a split is performed. Afterwards, a local EM is run in one dimension, and the two new components are projected back in the original space, giving a remarkable accuracy in describing the input data while maintaining a low computational burden. Nevertheless, all previous solutions are quite heuristic and do not follow any global optimally criteria. The drawback of the mentioned solutions is that even in the simpler case of one-dimensional data, the adopted split-based techniques use heuristic rules.

In this work, we strictly address the splitting problem, by proposing a global optimal algorithm, the 1-dimensional Gaussian Pair Mixture (GPM1D), to perform splits for GMMs estimation. Our technique is based on the explicit maximization of the likelihood of the data on a discrete set of tentative mixture models, achieved by a suitable discretization of the mixture parameters. Specifically, we create a table containing the parameters of the two mixture components (means, variances, weights) generated by means of the splitting operation, discretized with a determined quantization step. Then, we evaluate the EM log-likelihood of the input data corresponding to each combination of the discretized component parameters, which correspond to as many solutions. Finally, we choose the parameters that lead to the highest data likelihood. We compared our approach versus the original EM algorithm, both in terms of precision and computational burden.

III. An ML method to estimate a one-dimensional two-Gaussian mixture model

A. 2-Gaussian mixture model estimation

A general 1D mixture of K Gaussians is given by:

$$f(x) = \sum_{k=1}^{K} w_k \phi(x, m_k, \sigma_k), \quad \sum_{k=1}^{K} w_k = 1 \tag{1}$$

where w_k are the mixture weights, ϕ is a normalized Gaussian density function, m_k are the component means and σ_k their standard deviation. Given a set of N points $\{x_i\}_{i=1,\cdots,N}$, the goal is to estimate the mixture parameters w_k, m_k and σ_k.

The core of the proposed method consists in discretizing the space of parameters in a finite set of L points, each one corresponding to a particular instance of the mixture (hypothesis):

$$H = \{h_k^{(l)} = (w_k^{(l)}, m_k^{(l)}, \sigma_k^{(l)})\}, \quad l = 1, \cdots, L \tag{2}$$

Then, the log-likelihood of the data is computed for each hypothesis,

$$L^{(l)} = \sum_{i=1}^{N} log \left(\sum_{k=1}^{K} w_k^{(l)} \phi(x_i, m_k^{(l)}, \sigma_k^{(l)}) \right) \tag{3}$$

and the maximal one is selected as the best estimate of the mixture:

$$l^* = \underset{l}{\operatorname{argmax}}\, L^{(l)} \tag{4}$$

However, the number of hypotheses to evaluate grows exponentially with the number of components and partitions of the parameter set, and becomes impractical even for coarse discretizations. With $K = 2$, the parameter space has a dimension of 5 (i.e. w_1, m_1, m_2, σ_1 and σ_2), and a naive discretization may lead to a very high number of samples. For instance, if the range of each parameter is discretized in 10 points, the total number of hypotheses is 10^5, which is already too much for real time implementations. Fortunately, the parameters are not independent given the statistics of the whole data set. In the next section to derive a practical sampling of the space for 1D mixtures of two Gaussians.

B. Moments of a one-dimensional GMM

The first order moment (expected value) of a random variable x distributed according to (1) can be computed as:

$$\begin{aligned} m = E[x] &= \int x f(x) dx \\ &= \sum_{k=1}^{K} w_k \int x \phi(x, m_k, \sigma_k) dx = \sum_{k=1}^{K} w_k m_k \end{aligned} \tag{5}$$

To compute the second order central moment (variance) we first note that:

$$\begin{aligned} \sigma^2 &= E[(x-m)^2] = E[x^2 - 2mx + m^2] \\ &= E[x^2] - 2mE[x] + m^2 = E[x^2] - m^2 \end{aligned} \tag{6}$$

implies:

$$E[x^2] = \int x^2 f(x) dx = \sigma^2 + m^2$$

Analogously:

$$\int x^2 \phi(x, m_k, \sigma_k) dx = \sigma_k^2 + m_k^2.$$

Thus:

$$\sigma^2 + m^2 = E[x^2] = \int x^2 f(x) dx$$
$$= \sum_{k=1}^{K} w_k \int x^2 \phi(x, m_k, \sigma_k) dx = \sum_{k=1}^{K} w_k (\sigma_k^2 + m_k^2) \tag{7}$$

C. Parameter space of a two-gaussian 1-D mixture model: GPM1D

Given a Gaussian mixture model with two components and a single dimension, the mean (m) and variance (σ) of the whole mixture can be computed through (5) and (7):

$$m = w_1 m_1 + w_2 m_2 \tag{8}$$
$$\sigma^2 = w_1(\sigma_1^2 + m_1^2) + w_2(\sigma_2^2 + m_2^2) - m^2 \tag{9}$$

where $w_2 = 1 - w_1$, and $w_1 \in [0, 1]$.

Without loss of generality, we can normalize the data of the mixture to zero mean and unit variance:

$$0 = w_1 m_1 + (1 - w_1) m_2 \tag{10}$$
$$1 = w_1(\sigma_1^2 + m_1^2) + (1 - w_1)(\sigma_2^2 + m_2^2) \tag{11}$$

Given the five variables to estimate and the two above constrains, we have three algebraic degrees of freedom. Our strategy will consist in defining a suitable discretization for three of the variables, and the remaining two will be computed analytically from the previous constraints.

Although a three-dimensional space is more favorable for discretization than the original space, it still requires a large number of samples for good accuracy in the computations. Fortunately, many of the solution of a naive discretization are not feasible due to range contraints in the variables, e.g the variances cannot be negative, and the mixture weights are bound to the unit range. In fact, the cases of $w_1 = 0$ and $w_1 = 1$ are excluded because they correspond to a single Gaussian. This case can be verified by a prior statistical test, e.g. [15] . Therefore, we will consider $w_1, w_2 \in]0, 1[$. From (10) it is also easy to infer that:

$$m_1 < 0 \implies m_2 > 0$$
$$m_1 = 0 \implies m_2 = 0$$
$$m_2 < 0 \implies m_1 > 0$$

Without loss of generality, we can choose the order of the components so that $m_1 \leq 0$.

Let us first look at the case $m_1 < 0$, $m_2 > 0$. Using (10) in (11) we get:

$$m_2 \sigma_1^2 - m_1 \sigma_2^2 = (m_2 - m_1)(1 + m_1 m_2) \tag{12}$$

We can express σ_1^2 as a function of σ_2^2 and vice-versa:

$$\sigma_1^2 = \frac{(m_2 - m_1)(1 + m_1 m_2) + m_1 \sigma_2^2}{m_2} \tag{13}$$
$$\sigma_2^2 = \frac{(m_2 - m_1)(1 + m_1 m_2) - m_2 \sigma_1^2}{-m_1} \tag{14}$$

To solve these expressions (on the real numbers), the numerators of the right-hand side expressions must be nonnegative (note that $m_1 < 0$ and $m_2 > 0$):

$$(m_2 - m_1)(1 + m_1 m_2) + m_1 \sigma_2^2 \geq 0 \tag{15}$$
$$(m_2 - m_1)(1 + m_1 m_2) - m_2 \sigma_1^2 \geq 0 \tag{16}$$

which imposes the following constraints on σ_1^2 and σ_2^2:

$$\sigma_2^2 \leq \frac{(m_2 - m_1)(1 + m_1 m_2)}{-m_1} \tag{17}$$
$$\sigma_1^2 \leq \frac{(m_2 - m_1)(1 + m_1 m_2)}{m_2} \tag{18}$$

A solution exists if the numerator in the right hand side is non-negative, which, given the assumption of $m_1 < 0$ and $m_2 > 0$, implies that:

$$(m_2 - m_1)(1 + m_1 m_2) \geq 0 \implies m_1 m_2 \geq -1 \tag{19}$$

D. Discretization

There are many possible ways to discretize the parameter space of a 1-dimensional pair of Gaussians. In this paper we opted to discretize the space of each of the means m_1, m_2 in N_m hypotheses over a fixed range, and then discretize σ_2 in N_σ hypotheses over a range determined by (17). Let:

$$m_1^{(i)} = -i \frac{\Delta}{N_m},\ i = 1 \cdots N_m \tag{20}$$
$$m_2^{(j)} = j \frac{\Delta}{N_m},\ j = 1 \cdots N_m \tag{21}$$

where i and j index the discrete hypotheses of m_1 and m_2. For each combination of hypotheses $m_1^{(i)}$ and $m_2^{(j)}$, those that do not verify (19) would result in negative variances and can be excluded. From the remaining hypotheses of $m_1^{(i)}$ and $m_2^{(j)}$ we can analytically compute $w_1^{(ij)}$ using (10):

$$w_1^{(ij)} = \frac{m_2^{(i)}}{m_2^{(i)} - m_1^{(j)}} \tag{22}$$

Also, from (17) we can compute an upper bound on σ_2 for each valid hypotheses of $m_1^{(i)}$ and $m_2^{(j)}$:

$$\bar{\sigma_2}^{(ij)} = \sqrt{\frac{(m_1^{(i)} - m_2^{(j)})(1 + m_1^{(i)} m_2^{(j)})}{m_1^{(i)}}} \tag{23}$$

Now, σ_2 is discretized in N_σ steps:

$$\sigma_2^{(ijk)} = k \frac{\bar{\sigma_2}^{(ij)}}{N_\sigma} \tag{24}$$

Finally, the corresponding σ_1 can be computed analytically with (13):

$$\sigma_1^{(ijk)} = \sqrt{\frac{(m_2^{(j)} - m_1^{(i)})(1 + m_1^{(i)} m_2^{(j)}) + m_1^{(i)} (\sigma_2^{ijk})^2}{m_2^{(j)}}} \tag{25}$$

These steps are described in Algorithm III.1.

Algorithm III.1 GPM1D discretization rule

Require: data quantization step δ_w, number of quantization steps N_m

Ensure: new components w_1, w_2, m_1, m_2, σ_1, and σ_2

1: define quantization step δ_w
2: define the upper bound for $m_1 = 1/\sqrt{\delta_w}$
3: evaluate $m_1 \in \{-\bar{m}_1 + \delta_{m_1}, -\bar{m}_1 + 2\delta_{m_1}, \cdots, -\delta_{m_1}\}$
4: evaluate $m_2 = -\frac{w_1}{1-w_1} m_1$
5: fix the number of quantization steps N_δ for σ_2 and then the quantization step $\sigma_2^{min} = \sigma_2 / N_\delta$
6: evaluate $\sigma_2 = \left\{\sigma_2^{\text{min}}, 2 \cdot \sigma_2^{\text{min}}, \cdots, N_\sigma \cdot \sigma_2^{\text{min}}\right\}$
7: evaluate $w_2 = 1 - w_1$ $\{\forall$ quantized value$\}$
8: evaluate σ_1 as square root of (eq. 13): $\sigma_1^2 = \frac{(m_2 - m_1)(1 + m_1 m_2) + m_1 \sigma_2^2}{m_2}$ $\{\forall$ quantized value$\}$

IV. Validation and Discussion

A. Experimental set-up

We applied the proposed GPM1D method and the classical EM methods to the estimation of 1D Gaussian pair mixtures for different set sizes (number of data points). We fixed $\Delta = 3$ and tested different configurations for the number of mean and standard deviation hypotheses. The number of mean hypotheses N_m was in the set (10,15,20,25). The number of standard deviation hypotheses N_σ was in the set (3,6,9,12,15,18,21). Tab. I shows the combinations of mean and standard deviation hypotheses employed for the experimental validation. For the EM method, the default parameters of the Matlab R2015a statistics toolbox implementation were used.

The statistics of the log-likelihood, L2-distance and computation time distributions over 100 runs are presented in Tables II, III, IV and V, respectively, for datasets with 10^2, 10^3, 10^4 and 10^5 data points. We evaluated three parameters in order to compare the efficiency of GPM1D against EM: the L2-distance from the generated mixture - ground truth, the final log-likelihood, and the computational time. These are graphically shown in Fig. 1 (L2-distance), 2 (final log-likelihood), and 3 (computational time). Here, each of the four plots per picture represents the ML results using the four different input data sets. Each column is relative to a N_σ value $(3, 6, 9, 12, 15, 18, 21)$, grouped in the x-axis based on the number of N_m $(10, 15, 20, 25)$. The figures show the relative performance of the different configurations of GPM1D with respect to the EM baseline. Although for GPM1D we take into account each N_m-N_σ configuration (represented as the single columns in Fig. 1, 2, and 3), the EM values do not change (the EM is not affected by the GPM1D configuration), apart from the computational time. Therefore, we averaged the variant EM values, in order to compare them with the GPM1D output.

		std bins						
		3	6	9	12	15	18	21
mean bins	10	81	162	243	324	405	486	567
	15	198	396	594	792	990	1188	1386
	20	372	744	1116	1488	1860	2232	2604
	25	594	1188	1782	2376	2970	3524	4158

TABLE I

Number of hypotheses for each combination of mean bins with std bins. Used mean bins = [10,15,20,25]. Used std bins = [3,6,9,12,15,18,21]

B. Discussion

We can observe that for small data set sizes (from 100 to 1000 data points) there are configurations of GPM1D that outperform EM both in average fitness and in average computation time. These are represented as all the columns in Fig. 1, 2, and 3 standing below the red line representing the EM performance. Specifically, GPM1D gets better results for higher numbers of N_m-N_σ configuration, together with larger input data set. Tab. VI summarizes the configuration where GPM1D performs better than EM.

It is worthwhile noticing that for large data sets (from 10.000 to 100.000 data points), the configurations of GPM1D that present average results better that EM always require larger computation. Nevertheless, since GPM1D is easily parallelizable, the computation time can be significantly reduced in parallel implementations. Finally, for small datasets, GPM1D is more efficient than EM, for similar fitness values, even in sequential implementations.

V. Contextualization

The results confirm that the proposed GPM1D method provides an effective and robust solution to the split subproblem in incremental GMM estimation. Unlike heuristic strategies, GPM1D produces deterministic and globally optimal (up to discretization) splits, thereby reducing sensitivity to initialization and mitigating the convergence issues typically observed in EM-based refinements. This makes it particularly attractive for large-scale or time-sensitive applications, where reproducibility and computational efficiency are critical.

GPM1D does not aim to replace complete mixture estimation algorithms, but rather to provide a modular component that addresses a critical ill-posed step in these methods. Its integration into existing frameworks is straightforward, since it only substitutes the initialization of new components after a split. A key advantage of GPM1D is its generality. Although the method operates in a one-dimensional projected subspace, its design makes it especially suitable as a building block in algorithms that require repeated Gaussian component splits in multidimensional spaces. Examples of application of GPM1D are (but not only) the work cited in sec. II, such as [19], [1], and [6], as well as other greedy or incremental mixture-learning procedures. In these settings, GPM1D can serve as

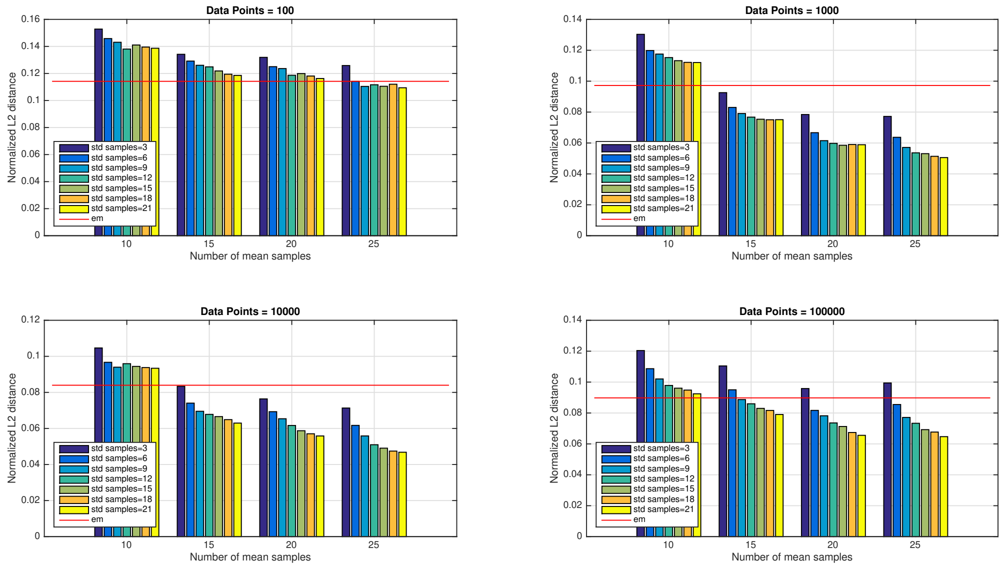

Fig. 1. Averages over 100 trials of the normalized L2-distance between the ground truth and estimated mixtures, using the classical EM and GPM1D, for several set sizes (10^2, 10^3, 10^4 and 10^5 data points).

100 points																								
Mean/std bins		Distance			Percentile distance				Time			Percentile time				Loglikelihood			Percentile loglikelihood			
mean	std	mean	std dev	median	2.5%	25%	75%	97.5%	mean	std dev	median	2.5%	25%	75%	97.5%	mean	std dev	median	2.5%	25%	75%	97.5%
10	3	0.153	0.310	0.041	0.0035	0.0199	0.114	1.20	0.00145	0.00045	0.00125	0.00109	0.00121	0.0016	0.0025	37.19	54.93	24.86	-37.7	-5.35	66.9	186.6
10	12	0.138	0.298	0.040	0.0044	0.0192	0.095	1.06	0.00452	0.00064	0.00431	0.00384	0.00418	0.0046	0.0068	39.22	55.64	26.01	-37.7	-3.34	68.3	186.8
10	21	0.139	0.294	0.042	0.0051	0.0213	0.095	1.04	0.00761	0.00100	0.00726	0.00651	0.00710	0.0080	0.0097	39.63	55.57	26.05	-37.7	-3.36	69.2	186.8
15	3	0.134	0.298	0.036	0.0034	0.0174	0.107	1.20	0.00298	0.00061	0.00276	0.00239	0.00265	0.0031	0.0051	39.52	58.20	25.92	-37.5	-5.08	69.8	201.8
15	12	0.125	0.274	0.041	0.0039	0.0186	0.076	1.01	0.01041	0.00139	0.00998	0.00889	0.00966	0.0108	0.0157	42.35	61.24	27.23	-37.5	-3.26	73.1	205.6
15	21	0.119	0.272	0.034	0.0027	0.0164	0.083	1.08	0.01860	0.00356	0.01798	0.01530	0.01727	0.0190	0.0280	42.90	61.93	27.56	-37.5	-3.04	74.6	205.6
20	3	0.132	0.304	0.034	0.0045	0.0188	0.075	1.28	0.00548	0.00127	0.00510	0.00430	0.00478	0.0058	0.0081	40.65	59.58	25.08	-37.5	-4.71	71.2	209.9
20	12	0.119	0.282	0.038	0.0051	0.0158	0.072	1.00	0.02039	0.00249	0.02010	0.01682	0.01898	0.0214	0.0289	42.88	61.08	27.82	-36.9	-3.00	74.1	210.1
20	21	0.116	0.279	0.034	0.0045	0.0202	0.065	1.04	0.03373	0.00373	0.03358	0.02884	0.03217	0.0347	0.0420	43.43	62.11	28.07	-37.2	-2.61	74.7	210.1
25	3	0.126	0.292	0.033	0.0015	0.0182	0.076	1.24	0.00812	0.00121	0.00777	0.00677	0.00743	0.0085	0.0121	41.49	60.16	25.92	-37.5	-4.67	71.5	204.1
25	12	0.112	0.265	0.036	0.0037	0.0179	0.068	1.00	0.03140	0.00323	0.03123	0.02652	0.02914	0.0337	0.0391	44.31	62.45	27.88	-35.9	-2.58	73.4	205.8
25	21	0.109	0.264	0.035	0.0018	0.0143	0.062	1.05	0.05470	0.00464	0.05555	0.04578	0.05098	0.0582	0.0638	44.87	63.33	28.03	-35.4	-2.87	74.4	206.1
EM		0.114	0.000	0.029	0.0008	0.0149	0.073	1.06	0.03042	0.01444	0.03876	0.00485	0.01748	0.0427	0.0475	41.89	57.28	28.36	-37.5	-2.65	73.0	181.1

TABLE II

STATISTICS OVER 100 RUNS OF THE LOG-LIKELIHOOD, L2-DISTANCE AND COMPUTATION TIME FOR GPM1D WITH DIFFERENT NUMBER OF HYPOTHESES OF MEAN AND STANDARD DEVIATION. 100 DATA POINTS.

1.000 points																						
Mean/std bins		Distance			Percentile distance				Time			Percentile time				Loglikelihood			Percentile loglikelihood			
mean	std	mean	std dev	median	2.5%	25%	75%	97.5%	mean	std dev	median	2.5%	25%	75%	97.5%	mean	std dev	median	2.5%	25%	75%	97.5%
10	3	0.130	0.278	0.029	0.0004	0.0047	0.119	1.23	0.0065	0.0020	0.0058	0.0052	0.0055	0.007	0.012	395.44	644.17	241.80	-439.1	-40.37	676.9	1801
10	12	0.115	0.275	0.016	0.0004	0.0049	0.092	1.27	0.0239	0.0040	0.0232	0.0196	0.0214	0.025	0.036	416.86	652.75	250.34	-434.9	-6.35	712.6	1812
10	21	0.112	0.275	0.017	0.0003	0.0036	0.094	1.29	0.0509	0.0224	0.0438	0.0357	0.0400	0.050	0.113	420.15	653.46	251.99	-434.6	-3.05	719.2	1812
15	3	0.093	0.250	0.013	0.0003	0.0038	0.059	0.90	0.0153	0.0043	0.0140	0.0122	0.0131	0.015	0.030	433.39	679.36	263.14	-437.5	-10.07	707.3	1986
15	12	0.077	0.243	0.008	0.0003	0.0026	0.033	0.90	0.0675	0.0242	0.0602	0.0498	0.0555	0.071	0.136	456.93	693.85	265.33	-423.6	4.20	794.0	1985
15	21	0.075	0.244	0.007	0.0003	0.0028	0.036	0.88	0.0972	0.0115	0.0949	0.0852	0.0902	0.099	0.136	460.13	695.65	272.05	-423.3	4.42	792.0	1986
20	3	0.078	0.225	0.013	0.0002	0.0026	0.055	0.56	0.0275	0.0037	0.0271	0.0226	0.0248	0.029	0.039	447.46	698.10	275.75	-426.9	-2.48	784.3	2403
20	12	0.060	0.218	0.006	0.0002	0.0022	0.022	0.53	0.1120	0.0190	0.1064	0.0946	0.1010	0.114	0.182	472.00	715.64	277.29	-424.3	4.60	819.7	2507
20	21	0.059	0.215	0.005	0.0001	0.0022	0.022	0.52	0.1862	0.0246	0.1781	0.1617	0.1717	0.191	0.263	475.66	720.55	276.46	-424.5	4.66	818.5	2516
25	3	0.077	0.227	0.007	0.0003	0.0027	0.057	0.56	0.0423	0.0053	0.0415	0.0355	0.0393	0.044	0.057	453.52	704.87	280.50	-422.6	0.69	808.0	2485
25	12	0.054	0.209	0.005	0.0001	0.0029	0.013	0.54	0.1695	0.0170	0.1657	0.1493	0.1592	0.174	0.215	479.01	724.83	299.87	-424.9	4.79	843.2	2503
25	21	0.051	0.203	0.005	0.0002	0.0026	0.016	0.51	0.2810	0.0352	0.2728	0.2568	0.2664	0.282	0.405	483.50	729.70	299.31	-422.6	4.78	843.4	2500
EM		0.097	0.000	0.008	0.0003	0.0029	0.057	0.99	0.0402	0.0226	0.0467	0.0042	0.0250	0.054	0.082	431.66	605.12	298.11	-422.0	4.37	804.3	1972

TABLE III

STATISTICS OVER 1.000 RUNS OF THE LOG-LIKELIHOOD, L2-DISTANCE AND COMPUTATION TIME FOR GPM1D WITH DIFFERENT NUMBER OF HYPOTHESES OF MEAN AND STANDARD DEVIATION. 1000 DATA POINTS.

a principled replacement for ad hoc or heuristic split rules, improving both the reliability and efficiency of the overall estimation pipeline.

Moreover, the computational profile of GPM1D aligns well with the requirements of such frameworks. The algorithm is non-iterative, parallelizable, and allows explicit control over the trade-off between accuracy and runtime. Once embedded within iterative methods like SMEM, it can significantly reduce the number of re-estimation steps required to reach convergence, thereby enhancing scalability in high-dimensional or data-intensive problems.

Nevertheless, some limitations should be acknowledged. The reliance on a one-dimensional projection may lose information in cases where variance is distributed across multiple

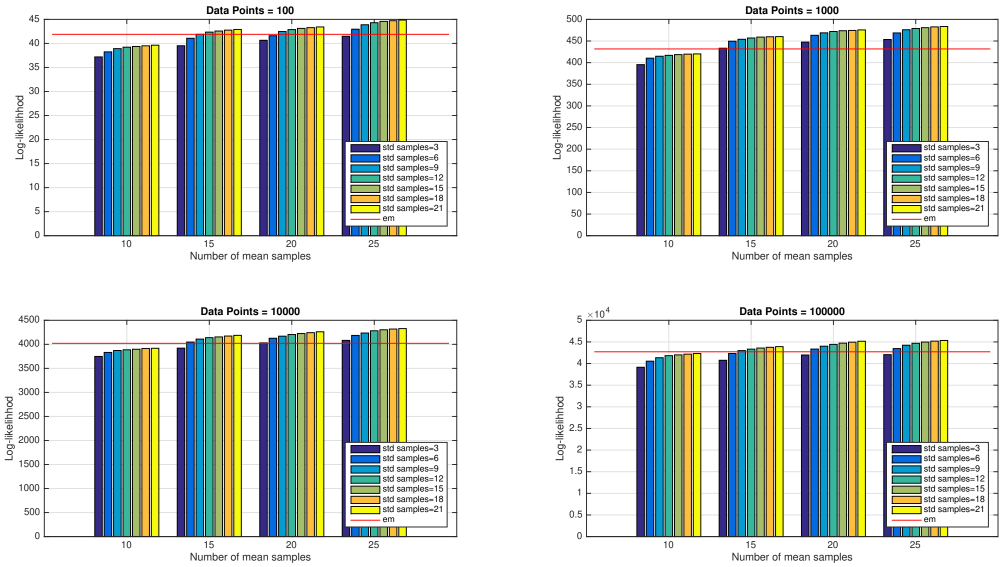

Fig. 2. Averages over 100 trials of the data log-likelihood on the mixtures computed by the classical EM and GPM1D, for several set sizes (10^2, 10^3, 10^4 and 10^5 data points).

10.000 points																				
Mean/std bins		Distance			Percentile distance				Time			Percentile time				Loglikelihood			Percentile loglikelihood			
mean	std	mean	std dev	median	2.5%	25%	75%	97.5%	mean	std dev	median	2.5%	25%	75%	97.5%	mean	std dev	median	2.5%	25%	75%	97.5%
10	3	0.105	0.260	0.011	0.0001	0.0029	0.061	0.89	0.043	0.011	0.042	0.030	0.033	0.051	0.066	3748	5704	2281	-3038.7	-377.7	6308	18442
10	12	0.096	0.251	0.006	0.0001	0.0015	0.052	0.92	0.147	0.018	0.145	0.123	0.133	0.159	0.185	3886	5782	2385	-2956.2	-276.0	6973	18461
10	21	0.093	0.249	0.007	0.0001	0.0010	0.056	0.92	0.262	0.048	0.245	0.211	0.227	0.286	0.381	3918	5786	2390	-2975.9	-275.6	7008	18461
15	3	0.083	0.252	0.008	0.0000	0.0019	0.037	0.89	0.103	0.025	0.098	0.073	0.087	0.111	0.163	3921	5910	2502	-3035.9	-274.9	6578	18895
15	12	0.068	0.224	0.004	0.0001	0.0008	0.029	0.56	0.408	0.091	0.375	0.307	0.330	0.490	0.601	4139	6182	2555	-2965.0	-274.9	7031	19086
15	21	0.063	0.218	0.004	0.0000	0.0007	0.028	0.52	0.605	0.089	0.570	0.511	0.540	0.638	0.832	4187	6277	2562	-2956.6	-273.8	7036	19077
20	3	0.076	0.246	0.005	0.0000	0.0010	0.030	0.87	0.179	0.033	0.168	0.139	0.153	0.202	0.255	4031	5954	2660	-3038.7	-275.1	6813	19415
20	12	0.062	0.227	0.002	0.0001	0.0007	0.017	0.73	0.723	0.125	0.692	0.559	0.639	0.796	1.014	4204	6154	2704	-2956.2	-220.2	7027	19459
20	21	0.056	0.220	0.002	0.0000	0.0005	0.016	0.66	1.191	0.235	1.134	0.982	1.037	1.242	2.062	4261	6263	2705	-2904.0	-272.0	7042	19461
25	3	0.071	0.239	0.005	0.0000	0.0007	0.027	0.84	0.271	0.057	0.250	0.214	0.232	0.288	0.441	4082	6044	2647	-3024.0	-273.2	6911	19691
25	12	0.051	0.211	0.002	0.0000	0.0004	0.014	0.61	1.108	0.224	1.024	0.884	0.962	1.173	1.814	4282	6286	2677	-2928.6	-254.6	7121	19701
25	21	0.047	0.205	0.001	0.0001	0.0005	0.013	0.60	2.104	0.278	2.103	1.632	2.014	2.188	2.980	4327	6342	2681	-2938.9	-271.0	7142	19699
EM		0.084	0.000	0.003	0.0001	0.0008	0.023	0.95	0.138	0.078	0.156	0.011	0.071	0.195	0.252	4021	5762	2656	-3173.2	-273.8	7013	19187

TABLE IV

STATISTICS OVER 10.000 RUNS OF THE LOG-LIKELIHOOD, L2-DISTANCE AND COMPUTATION TIME FOR GPM1D WITH DIFFERENT NUMBER OF HYPOTHESES OF MEAN AND STANDARD DEVIATION. 10.000 DATA POINTS.

100.000 points																						
Mean/std bins		Distance			Percentile distance				Time			Percentile time				Loglikelihood			Percentile loglikelihood			
mean	std	mean	std dev	median	2.5%	25%	75%	97.5%	mean	std dev	median	2.5%	25%	75%	97.5%	mean	std dev	median	2.5%	25%	75%	97.5%
10	3	0.105	0.260	0.011	0.0001	0.0029	0.061	0.89	0.043	0.011	0.042	0.030	0.033	0.051	0.066	3748	5704	2281	-3038.7	-377.7	6308	18442
10	12	0.096	0.251	0.006	0.0001	0.0015	0.052	0.92	0.147	0.018	0.145	0.123	0.133	0.159	0.185	3886	5782	2385	-2956.2	-276.0	6973	18461
10	21	0.093	0.249	0.007	0.0001	0.0010	0.056	0.92	0.262	0.048	0.245	0.211	0.227	0.286	0.381	3918	5786	2390	-2975.9	-275.6	7008	18461
15	3	0.083	0.252	0.008	0.0000	0.0019	0.037	0.89	0.103	0.025	0.098	0.073	0.087	0.111	0.163	3921	5910	2502	-3035.9	-274.9	6578	18895
15	12	0.068	0.224	0.004	0.0001	0.0008	0.029	0.56	0.408	0.091	0.375	0.307	0.330	0.490	0.601	4139	6182	2555	-2965.0	-274.9	7031	19086
15	21	0.063	0.218	0.004	0.0000	0.0007	0.028	0.52	0.605	0.089	0.570	0.511	0.540	0.638	0.832	4187	6277	2562	-2956.6	-273.8	7036	19077
20	3	0.076	0.246	0.005	0.0000	0.0010	0.030	0.87	0.179	0.033	0.168	0.139	0.153	0.202	0.255	4031	5954	2660	-3038.7	-275.1	6813	19415
20	12	0.062	0.227	0.002	0.0001	0.0007	0.017	0.73	0.723	0.125	0.692	0.559	0.639	0.796	1.014	4204	6154	2704	-2956.2	-220.2	7027	19459
20	21	0.056	0.220	0.002	0.0000	0.0005	0.016	0.66	1.191	0.235	1.134	0.982	1.037	1.242	2.062	4261	6263	2705	-2904.0	-272.0	7042	19461
25	3	0.071	0.239	0.005	0.0000	0.0007	0.027	0.84	0.271	0.057	0.250	0.214	0.232	0.288	0.441	4082	6044	2647	-3024.0	-273.2	6911	19691
25	12	0.051	0.211	0.002	0.0000	0.0004	0.014	0.61	1.108	0.224	1.024	0.884	0.962	1.173	1.814	4282	6286	2677	-2928.6	-254.6	7121	19701
25	21	0.047	0.205	0.001	0.0001	0.0005	0.013	0.60	2.104	0.278	2.103	1.632	2.014	2.188	2.980	4327	6342	2681	-2938.9	-271.0	7142	19699
EM		0.084	0.000	0.003	0.0001	0.0008	0.023	0.95	0.138	0.078	0.156	0.011	0.071	0.195	0.252	4021	5762	2656	-3173.2	-273.8	7013	19187

TABLE V

STATISTICS OVER 100.000 RUNS OF THE LOG-LIKELIHOOD, L2-DISTANCE AND COMPUTATION TIME FOR GPM1D WITH DIFFERENT NUMBER OF HYPOTHESES OF MEAN AND STANDARD DEVIATION. 100.000 DATA POINTS.

directions, and discretization introduces a runtime–accuracy trade-off. Further empirical validation on real-world high-dimensional data will be important to assess the robustness of the approach. In summary, GPM1D offers a principled, efficient, and general-purpose solution to the Gaussian component split problem, making it a natural candidate for integration into advanced GMM estimation algorithms such as SMEM and related approaches.

VI. Conclusions

We introduced GPM1D, a deterministic method to address the ill-posed problem of splitting Gaussian components in mixture models. By discretizing the parameter space into a finite

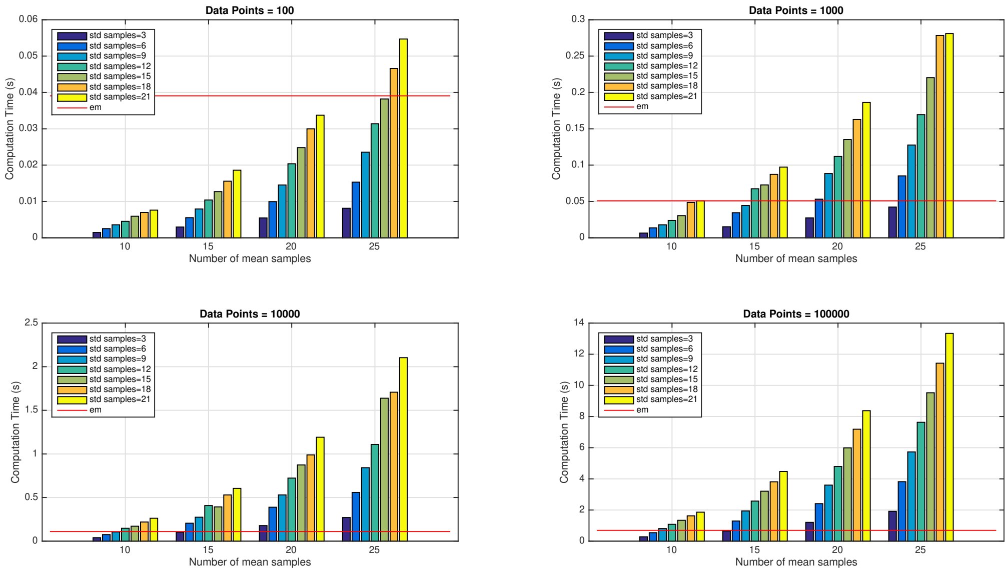

Fig. 3. Averages over 100 trials of the computation time of the classical EM and GPM1D, for several set sizes (10^2, 10^3, 10^4 and 10^5 data points).

100 points		std bins						
		3	6	9	12	15	18	21
param.		L2-LL-TM	L2-LL-TM	L2-LL-TM	L2-LL-TM	L2-LL-TM	L2-LL-TM	L2-LL-TM
mean bins	10	EM-EM-EM	EM-EM-EM	EM-EM-EM	EM-EM-EM	EM-EM-EM	EM-EM-EM	EM-EM-EM
	15	EM-EM-EM	EM-EM-EM	EM-EM-EM	EM-**GP**-EM	EM-**GP**-EM	EM-**GP**-EM	EM-**GP**-EM
	20	EM-EM-EM	EM-EM-EM	EM-**GP**-EM	EM-**GP**-EM	EM-**GP**-EM	EM-**GP**-EM	EM-**GP**-EM
	25	EM-EM-EM	EM-**GP**-EM	EM-**GP**-EM	**GP-GP**-EM	**GP-GP**-EM	**GP-GP-GP**	**GP-GP-GP**

TABLE VI

Best algorithm result with respect to: L2 Distance (L2), log-likelihood (LL), and computation time (TM). The two algorithms in the comparison are the Expectation Maximization (EM) and our GPM1D (GP). For instance, std. bins = 15 and mean bins = 25, GPM1D outperformed EM in L2 distance and log-likelihood, while EM has the best performance in computation time (GP-GP-EM). Used mean bins = [10,15,20,25]. Used std bins = [3,6,9,12,15,18,21]. As long as the values for mean bins and st. bins increase, generally GPM1D outperforms EM in all the cited areas.

grid, the method converts the split problem into a tractable optimization task. Experimental comparisons with EM show that even a non-optimized MATLAB implementation achieves competitive results, while the algorithm's simplicity makes it inherently parallelizable. Rather than a full estimation framework, GPM1D provides a modular split function that can be seamlessly integrated into incremental and/or greedy methods. By replacing heuristic split rules with GPM1D, these approaches can gain improvements in convergence speed, stability, and reproducibility. Future work will focus on large-scale experiments, adaptive discretization, and a parallelized implementation to exploit the method's scalability.

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